United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam Spring 2025

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: **30** Duration: 1 hour 30 Minutes

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. (a) Show the corresponding inputs and outputs from **Lexical analyzer** to **Code Optimizer** phase of compiler based on the following expression where x and z are "**float**" type and others are "**int**" type. $P = Z * (2 + X)^\wedge Y$ 4

   (b) Imagine you are developing a cross-platform banking application in **Java.** You wrote the code on a Windows machine, and now it needs to run on a Linux server. Explain how Java's **hybrid compilation process** ensures that your banking application can run on both Windows and Linux without modification. 3

2. Write down the three address representation of the following expression. 4

$$z = x + y - (r/2 + q)/2^\wedge 4^\wedge 5 - a * (3/2)^\wedge(1^\wedge t)$$

t12

3. Draw Syntax Tree for the following expression. 4

$$x = (b * 2/5^\wedge 6)^\wedge(g * (5^\wedge d)^\wedge 2 \,/\, 1)\ ^\wedge 3 - r * 2/k$$

root -

4. Prove that the following expression "111010011" can be represent by the grammar given below. 4

   S → X | Y | Z
   X → D10D
   D → 0Z | 1D | 1Y | ε
   Y → 0Y | 0
   Z → 1Z | 1 | ε

derived

5. Convert the following infix expression to a prefix expression. 4

$$3 + y^\wedge 2^\wedge q^\wedge p/(r + p - 2^\wedge 3 * 5) - (a + 3)^\wedge 1^\wedge 3$$

- + 3/^y^2^qp - +rp*^235 ^+a3 ^13

6. Convert the following infix expression to a postfix expression. 4

$$2 * (3 * 4^\wedge(5 - 1)^\wedge d) + 5 - (2/4 * 6 + p\,/b * 3)$$

23451-d^^**5+24/6*pb/3*+-

7. Evaluate the following postfix expression. 3

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6 3 * 2 + 4 / 5 2 - 3 ^ 8 4 / + 7 2 * 3 - 5 1 + / - +
```

32.12